Boolean Algebra Solutions – Senior Division

1. **00-01 C3 Boolean Algebra**

$\overline{\overline{A+B}+\overline{AB}}+\overline{A\oplus B}=(A+B)(AB)+AB+\overline{A}\,\overline{B}=$
$AAB+ABB+AB+\overline{A}\,\overline{B}=AB+\overline{A}\,\overline{B}$

$AB+\overline{A}\,\overline{B}$

2. **01-02 C2 Boolean Algebra**

$(\overline{AB}C)(\overline{A}C)=(\overline{A}+\overline{B})C\overline{A}C=\overline{A}C+\overline{A}\,\overline{B}C=\overline{A}C$

$\overline{A}C$

3. **01-02 C2 Boolean Algebra**

The Boolean expression is: $AB\oplus\overline{C+(CB+A)}$

Let X = AB, Y = CB, Z = CB+A, W = C + (CB + A)

A	B	C	X	Y	Z	W	NOT W	X⊕NOT W
0	0	0	0	0	0	0	1	1
0	0	1	0	0	0	1	0	0
0	1	0	0	0	0	0	1	1
0	1	1	0	1	1	1	0	0
1	0	0	0	0	1	1	0	0
1	0	1	0	0	1	1	0	0
1	1	0	1	0	1	1	0	1
1	1	1	1	1	1	1	0	1

(0,0,0), (0,1,0), (1,1,0), (1,1,1)

4. **01-02 C3 Boolean Algebra**

$(A+B)\oplus AB=(A+B)(\overline{AB})+\overline{A+B}AB=$
$(A+B)(\overline{A}+\overline{B})+\overline{A}\,\overline{B}AB=(A+B)(\overline{A}+\overline{B})=$
$A\overline{A}+A\overline{B}+B\overline{A}+B\overline{B}=A\overline{B}+B\overline{A}=A\oplus B$

$A\oplus B$

5. **02-03 C2 Boolean Algebra**

$\overline{B(\overline{A}+\overline{B})+A(\overline{\overline{B}\,\overline{A}+\overline{B}})}=\overline{\overline{A}B+B\overline{B}+A(\overline{B}\,\overline{\overline{A}}\,\overline{\overline{B}})}=$
$\overline{\overline{A}B+AB(\overline{\overline{B}\,\overline{A}})}=\overline{\overline{A}B+AB(\overline{B}+A)}=$
$\overline{\overline{A}B+AB\overline{B}+ABA)}=\overline{\overline{A}B+AB}=\overline{B(\overline{A}+A)}=\overline{B}$

C and E

Choice a. simplifies to $\overline{\overline{A}B+AB\overline{B}\,\overline{\overline{A}}}=\overline{\overline{A}B+AB(\overline{B}+A)}=\overline{\overline{A}B+AB\overline{B}+AAB}=\overline{\overline{A}B+AB}=\overline{B}$.

Choice c. simplifies to $\overline{A}B+AB=(\overline{A}+A)B=B$.

Choice d. simplifies to $\overline{\overline{A}B+AB}=\overline{(\overline{A}+A)B}=\overline{B}$.

Therefore, options c and e are NOT equivalent.

6. **02-03 C2 Boolean Algebra**

A AND NOT B XOR B AND (NOT A OR C) (0,0,0), (0,0,1), (1,1,0)

A	**B**	**C**	$A\overline{B}$	$\overline{A}+C$	$B(\overline{A}+C)$	**XOR**
0	**0**	**0**	**0**	**1**	**0**	**0**
0	**0**	**1**	**0**	**1**	**0**	**0**
0	**1**	**0**	**0**	**1**	**1**	**1**
0	**1**	**1**	**0**	**1**	**1**	**1**
1	**0**	**0**	**1**	**0**	**0**	**1**
1	**0**	**1**	**1**	**1**	**0**	**1**
1	**1**	**0**	**0**	**0**	**0**	**0**
1	**1**	**1**	**0**	**1**	**1**	**1**

7. **02-03 C3 Boolean Algebra**

$\overline{A}B + B(A+\overline{C}) + (\overline{C}+B)(\overline{A}+\overline{B}) =$

$\overline{A}B + BA + B\overline{C} + \overline{C}\,\overline{A} + B\overline{A} + \overline{C}\,\overline{B} + B\overline{B} =$ (*,1,*), (*,0,0) or

$(\overline{A}+A)B + \overline{C}(B+\overline{B}+\overline{A}) = B+\overline{C}$

A = *. If B = 1, then C = *. If B = 0, then C = 0. (0,0,0), (0,1,0), (0,1,1)

The triples are: (0,0,0), (0,1,0), (0,1,1), (1,0,0), (1,1,0) and (1,1,1). (1,0,0), (1,1,0), (1,1,1)

8. **03-04 C2 Boolean Algebra**

The expression is: $\overline{A}(AB+\overline{A})$ (0,0) and (0,1)

A	B	$\overline{A}$	AB	$AB+\overline{A}$	$\overline{A}(AB+\overline{A})$
0	0	1	0	1	1
0	1	1	0	1	1
1	0	0	0	0	0
1	1	0	1	1	0

9. **03-04 C2 Boolean Algebra**

$\overline{A(\overline{B}+C)} + \overline{A\overline{B}+C} = \overline{A} + \overline{\overline{B}+C} + \overline{A\overline{B}}\;\overline{C} =$ $\overline{A}+B\overline{C}$

$\overline{A} + \overline{\overline{B}}\;\overline{C} + (\overline{A}+\overline{\overline{B}})\overline{C} = \overline{A} + B\overline{C} + \overline{A}\,\overline{C} + B\overline{C} =$

$\overline{A}(1+C) + B\overline{C} = \overline{A} + B\overline{C}$

10. **03-04 C3 Boolean Algebra**

$A(\overline{(A+\overline{B})(B+C)} + B + C) = A(\overline{A+\overline{B}} + \overline{B+C} + B + C) =$ A

$A(\overline{A+\overline{B}} + 1) = A(1) = A$

11. **04-05 C2 Boolean Algebra**

$\overline{A+(B+\overline{C})}+\ \overline{B}(\overline{A+C})\ =\ \overline{A}\overline{B+\overline{C}}\ +\overline{B}*\overline{A}*\overline{C}\ =$ $\overline{A+B}$

$\overline{A}*\overline{B}*\overline{\overline{C}}\ +\overline{B}*\overline{A}*\overline{C}\ =\ \overline{A}\overline{B}(C+\overline{C})=\ \overline{A}\overline{B}=\overline{A+B}$

12. **04-05 C2 Boolean Algebra**

The truth table for $A(B\oplus C)+\overline{AB}$ has 7 of the 8 that make it TRUE 7

A	B	C	$B\oplus C$	$A(B\oplus C)$	$\overline{AB}$	$+$
0	0	0	0	0	1	1
0	0	1	1	0	1	1
0	1	0	1	0	1	1
0	1	1	0	0	1	1
1	0	0	0	0	1	1
1	0	1	1	1	1	1
1	1	0	1	1	0	1
1	1	1	0	0	0	0

13. **04-05 C3 Boolean Algebra**

$\overline{\overline{\overline{A}B+C}+\overline{\overline{B}(A+\overline{C})}}=\ \overline{\overline{A}B+C}*\overline{\overline{B}(A+\overline{C})}=$ $A\overline{B}C$

$(\overline{A}B+C)(A\overline{B}+\overline{B}\overline{C})=\ \overline{A}BA\overline{B}+\overline{A}B\overline{B}\overline{C}+A\overline{B}C+\overline{B}\overline{C}C=A\overline{B}C$

14. **05-06 C2 Boolean Algebra**

$\overline{A(\overline{B}+\overline{C})}+(A\oplus\overline{BC})=\ \overline{\overline{A}}+\overline{\overline{B}+\overline{C}}+\overline{A}\overline{BC}+A\overline{\overline{BC}}=$ 0 or NONE

$A+BC+\overline{A}(\overline{B}+\overline{C})+ABC\ =A+BC+\overline{AB}+\overline{A}\,\overline{C}+ABC$

To be FALSE all the terms must be FALSE. A must = 0. B and C must also = 0 to make the second term FALSE. This makes the third term TRUE. This is a contraction and the expression can never be FALSE.

15. **05-06 C2 Boolean Algebra**

$\overline{\overline{\overline{A\overline{B}(1+\overline{C})}+C(\overline{A}+B)}+\overline{A}(B\overline{C}+\overline{A})}=$ AB

$\overline{\overline{A\overline{B}}+C(\overline{A}+B)}\,\overline{A}(B\overline{C}+\overline{A})=(\overline{A\overline{B}}+C(\overline{A}+B))(\overline{\overline{A}}+(\overline{B\overline{C}+\overline{A}}))=$

$(\overline{A}+B+\overline{A}C+BC)(A+\overline{B\overline{C}}\,\overline{A})=(\overline{A}(1+C)+B(1+C))(A+A(\overline{B}+C))=$

$(\overline{A}+B)(A+(B+\overline{C})A)=(\overline{A}+B)(A(1+B+\overline{C})=$

$(\overline{A}+B)(A)=A\overline{A}+AB\ =AB$

16. **05-06 C3 Boolean Algebra**

$(A+B\overline{C})(\overline{A}\,\overline{B}+C)=A\overline{A}\,\overline{B}+AC+B\overline{C}\,\overline{A}\,\overline{B}+B\overline{C}C=0+AC+0+0=AC$ AC

17. **06-07 C2 Boolean Algebra**

$(\overline{\overline{A}B+AC})(\overline{\overline{C}+B})A=(\overline{\overline{A}B})(\overline{AC})(\overline{\overline{C}}\,\overline{B}A)=(\overline{\overline{A}}+\overline{B})(\overline{A}+\overline{C})(A\overline{B}C)=$ 0

$(A\overline{A}+A\overline{C}+\overline{A}\,\overline{B}+\overline{B}\,\overline{C})(A\overline{B}C)=$

$A\overline{A}A\overline{B}C+A\overline{C}A\overline{B}C+\overline{A}\,\overline{B}A\overline{B}C+\overline{B}\,\overline{C}A\overline{B}C=0$

18. **06-07 C2 Boolean Algebra**

$(\overline{A}\oplus B)(\overline{B}\oplus C)(\overline{C}\oplus\overline{A})=(\overline{A}\,\overline{B}+AB)(\overline{B}\,\overline{C}+BC)(\overline{A}C+A\overline{C})=$ 0

$(\overline{A}\,\overline{B}\,\overline{B}\,\overline{C}+\overline{A}\,\overline{B}BC+AB\overline{B}\,\overline{C}+ABBC)(\overline{A}C+A\overline{C})=$

$(\overline{A}\,\overline{B}\,\overline{C}+ABC)(\overline{A}C+A\overline{C})=$

$\overline{A}\,\overline{B}\,\overline{C}\overline{A}C+\overline{A}\,\overline{B}\,\overline{C}A\overline{C}+ABC\overline{A}C+ABCA\overline{C}=0$

19. **06-07 C3 Boolean Algebra**

$AB(C+A)+\overline{B}(A+BC)=ABC+ABA+A\overline{B}+\overline{B}BC=$ A

$ABC+AB+A\overline{B}=AB(C+1)+A\overline{B}=AB+A\overline{B}=A(B+\overline{B})=A$

20. **07-08 C2 Boolean Algebra**

$\overline{\overline{A(\overline{B}+C)}+B(AC+B)}=\overline{\overline{A(\overline{B}+C)}}\,\overline{B(AC+B)}=$ $A\overline{B}$

$=(A\overline{B}+AC)(\overline{B}+(\overline{A}+\overline{C})\overline{B})=(A\overline{B}+AC)(\overline{B}+\overline{A}\,\overline{B}+\overline{B}\,\overline{C})$

$=(A\overline{B}+AC)(\overline{B}(1+\overline{A}+\overline{C}))=$

$=(A\overline{B}+AC)(\overline{B})=A\overline{B}\,\overline{B}+A\overline{B}C=A\overline{B}(1+C)=A\overline{B}$

21. **07-08 C2 Boolean Algebra**

The last column yields (0,0,*), (1,0,0) and (1,1,1) 4

A	B	C	$\overline{B}$	$\overline{C}$	$B\oplus\overline{C}$	$A(B\oplus\overline{C})$	AC	$AC\oplus\overline{B}$	$\overline{B}(AC\oplus\overline{B})$	+
0	0	0	1	1	1	0	0	1	1	1
0	0	1	1	0	0	0	0	1	1	1
0	1	0	0	1	0	0	0	0	0	0
0	1	1	0	0	1	0	0	0	0	0
1	0	0	1	1	1	1	0	1	1	1
1	0	1	1	0	0	0	1	0	0	0
1	1	0	0	1	0	0	0	0	0	0
1	1	1	0	0	1	1	1	1	0	1

22. **07-08 C3 Boolean Algebra**

$(A+BC)(\overline{A}\overline{B}+C)(\overline{A}+\overline{B}C)(A\overline{B}+\overline{C})$ $A\overline{B}C$

$=(A\overline{A}\overline{B}+AC+\overline{A}B\overline{B}C+BC)(\overline{A}\overline{B}+C)(\overline{A}+\overline{B}C)(A\overline{B}+\overline{C})$

$=(AC+BC)(\overline{A}+\overline{B}C)(A\overline{B}+\overline{C}) = (A\overline{A}C+A\overline{B}C+\overline{A}BC+B\overline{B}C)(A\overline{B}+\overline{C})$

$=(A\overline{B}C+\overline{A}BC)(A\overline{B}+\overline{C}) = A\overline{B}C+A\overline{B}C\overline{C}+A\overline{A}B\overline{B}C+ABC\overline{C} = A\overline{B}C$

23. **08-09 C2 Boolean Algebra**

$\overline{A}(B+A)+A\overline{B} = \overline{A}B+\overline{A}A+A\overline{B} = \overline{A}B+0+A\overline{B} = A\oplus B$ $A\oplus B$

24. **08-09 C2 Boolean Algebra**

$A(B+\overline{C})+\overline{BC}+\overline{A(BC+A)}$ 8

$=AB+A\overline{C}+\overline{B}+\overline{C}+\overline{A}+\overline{A}(\overline{B}+\overline{C})$

$=AB+A\overline{C}+\overline{B}+\overline{C}+\overline{A}+\overline{A}\overline{B}+\overline{A}\overline{C} = AB+\overline{B}+\overline{C}+\overline{A}$

If $B=1$, then $AB+\overline{B}+\overline{C}+\overline{A} = A*1+0+\overline{C}+\overline{A} = A+\overline{C}+\overline{A} = 1$,

so $A=*\wedge C=*$

If $B=0$, then $A*0+1+\overline{C}+\overline{A}=1$, so $A=*\wedge C=*$.

Hence (*,*,*) is the solution so all 8 ordered triples make it TRUE..

25. **08-09 C3 Boolean Algebra**

$\overline{\overline{A\overline{B}}+\overline{A\overline{C}}+(AB\overline{C}+A\overline{B}C)} = \overline{\overline{AB+AC}}\,\overline{(AB\overline{C}+A\overline{B}C)}$ (1,0,0)

$=(A\overline{B}+A\overline{C})(\overline{\overline{AB}\,\overline{C}}\;\overline{A\overline{B}C}) = (A\overline{B}+A\overline{C})(\overline{A}+\overline{B}+C)(\overline{A}+B+\overline{C})$

$=(A\overline{B}+A\overline{C})(\overline{A}+\overline{B}\,\overline{C}+BC) = A\,\overline{B}\,\overline{C}$

$\therefore A=1\wedge B=0\wedge C=0$

26. **09-10 C2 Boolean Algebra**

$(A+B)(B+\overline{A}C)+AB\overline{C}+B(\overline{A}+\overline{C})$ B

$= AB+A\overline{A}C+BB+\overline{A}BC+AB\overline{C}+\overline{A}B+B\overline{C} =$

$= AB+0+B+\overline{A}BC+AB\overline{C}+\overline{A}B+B\overline{C}$

$= AB+B+\overline{A}BC+AB\overline{C}+\overline{A}B+B\overline{C} = B(A+1+\overline{A}C+A\vec{C}+\overline{A}+\overline{C}) = B$

27. **09-10 C2 Boolean Algebra**

$\overline{A\overline{B}+C}+A(B+\overline{C})\oplus\overline{A}BC = \overline{(A\overline{B}+C)}+(A(B+\overline{C})\oplus\overline{A}BC)$ (0,0,1), (1,0,1)

$= \overline{A\overline{B}}\,\overline{C}+A(B+\overline{C})\overline{\overline{A}BC}+\overline{A(B+\overline{C})}\overline{A}BC$

$= (\overline{A}+B)\overline{C}+A(B+\overline{C})(A+\overline{B}+\overline{C})+(\overline{A}+(\overline{B+\overline{C}}))\overline{A}BC$

$= (\overline{A}\overline{C}+B\overline{C})+(AB+A\overline{C})(A+\overline{B}+\overline{C})+(\overline{A}+\overline{B}\overline{\overline{C}})\overline{A}BC$

$= (\overline{A}\overline{C}+B\overline{C}+AAB+AB\overline{B}+AB\overline{C}+AA\overline{C}+A\overline{B}\overline{C}+A\overline{C}\overline{C}+\overline{A}\overline{A}BC+\overline{A}B\overline{B}CC$

$= \overline{A}\overline{C}+B\overline{C}+AB+0+AB\overline{C}+A\overline{C}+A\overline{B}\overline{C}+A\overline{C}+\overline{A}BC+0$

$= \overline{C}(\overline{A}+B+AB+A+A\overline{B})+AB+\overline{A}BC$

$= \overline{C}+AB+\overline{A}BC$ So $\overline{C}+AB+\overline{A}BC=0 \Rightarrow C=1 \wedge AB+\overline{A}B=0$

$\Rightarrow B(A+\overline{A})=0 \Rightarrow B=0 \wedge A=*$ $\therefore$(0,0,1) and (1,0,1)

28. **09-10 C3 Boolean Algebra**

$\overline{(\overline{A+\overline{B}})(\overline{B+C})(\overline{A}B)}=0 \Rightarrow (\overline{A+\overline{B}})(\overline{B+C})(\overline{A}B)=1$ 0

$\Rightarrow \overline{A+\overline{B}}=1 \wedge \overline{B+C}=1 \wedge \overline{A}B=1 \Rightarrow \overline{A}B=1 \wedge \overline{B}\overline{C}=1 \wedge \overline{A}B=1$

$\Rightarrow (A=0 \wedge B=1) \wedge (B=0 \wedge C=0) \wedge (A=0 \wedge B=1)$, a contradiction

29. **10-11 C2 Boolean Algebra**

There are 5 ordered triples that make it TRUE: (0,1,0), (0,1,1), (1,0,0), (1,1,0), (1,1,1)

A	B	C	$\overline{A}$	$\overline{C}$	$B+\overline{C}$	$A(B+\overline{C})$	$\overline{A}B$	$A(B+\overline{C})\oplus\overline{A}B$
0	0	0	1	1	1	0	0	0
0	0	1	1	0	0	0	0	0
0	1	0	1	1	1	0	1	1
0	1	1	1	0	1	0	1	1
1	0	0	0	1	1	1	0	1
1	0	1	0	0	0	0	0	0
1	1	0	0	1	1	1	0	1
1	1	1	0	0	1	1	0	1

30. **10-11 C2 Boolean Algebra**

$A(\overline{A\overline{B}+BC})(\overline{\overline{A}+B}+C) = A(\overline{A\overline{B}}\,\overline{BC})(\overline{A}\overline{B}+C)$ 0

$= A((\overline{A}+B)(\overline{B}+\overline{C})(\overline{A}\overline{B}+C)$

$= (\overline{A}\overline{B}+\overline{A}\overline{C}+B\overline{B}+B\overline{C})(A\overline{A}\overline{B}+AC) = (\overline{A}\overline{B}+\overline{A}\overline{C}+B\overline{C})AC$

$= \overline{A}\overline{B}AC+\overline{A}\overline{C}AC+B\overline{C}AC=0$

31. **10-11 C3 Boolean Algebra**

$$\overline{(\overline{A}+B)(\overline{A}+B+\overline{C})+AB(\overline{A}+C)} = \overline{(\overline{A}+B)(\overline{A}+B+\overline{C})}\,\overline{AB(\overline{A}+C)} \qquad A\overline{B}$$

$$= (\overline{\overline{A}+B}+\overline{\overline{A}+B+\overline{C}})(\overline{AB}+\overline{\overline{A}+C}) = (A\overline{B}+A\overline{B}C)(\overline{A}+\overline{B}+A\overline{C})$$

$$= A\overline{B}\overline{A}+A\overline{B}\overline{B}+A\overline{B}A\overline{C}+A\overline{B}C\overline{A}+A\overline{B}C\overline{B}+A\overline{B}CA\overline{C} = A\overline{B}+A\overline{B}\overline{C}+A\overline{B}C$$

$$= A\overline{B}(1+\overline{C}+C) = A\overline{B}$$

32. **11-12 C3 Boolean Algebra**

$$\overline{(A+B)+(\overline{B}+C)}+\overline{A}(\overline{B+C}) = (\overline{A+B})(\overline{B+C})+\overline{A}(\overline{B}\,\overline{C}) \qquad \overline{A}\,\overline{B}\,\overline{C}$$

$$= (\overline{A}\,\overline{B})(B\,\overline{C})+\overline{A}\,\overline{B}\,\overline{C}$$

$$= \overline{A}\,\overline{B}\,B\,\overline{C}+\overline{A}\,\overline{B}\,\overline{C} = \overline{A}\,\overline{B}\,\overline{C}$$

33. **11-12 C3 Boolean Algebra**

$$B(\overline{A(\overline{B}+C)})(\overline{C}(\overline{AB})) = B(\overline{A}+(\overline{\overline{B}+C}))(\overline{C}(\overline{A}+\overline{B})) \qquad (0,1,0),\ (0,1,1)$$

$$= B(\overline{A}+B\overline{C})(\overline{C}(\overline{A}+\overline{B})) = (\overline{A}B+BB\overline{C})(\overline{A}\overline{C}+\overline{B}C)$$

$$= \overline{A}B\overline{A}C+\overline{A}B\overline{B}\overline{C}+\overline{A}BB\overline{C}\overline{C}+BB\overline{C}\overline{B}\overline{C}$$

$$= \overline{A}BC+\overline{A}B\overline{C} = \overline{A}B(C+\overline{C}) = \overline{A}B$$

$$\overline{A}B = 0 \Rightarrow A = 0 \wedge B = 1 \wedge C = * \qquad \therefore (0,\ 1,\ *)$$

34. **12-13 C3 Boolean Algebra**

$$\overline{\overline{(\overline{A+\overline{B}})(AB)}}+\overline{(A+B)(\overline{\overline{A}\overline{B}})} = \overline{(\overline{A}\overline{B})(AB)}+\overline{(A+B)(A+\overline{B})} \qquad 1$$

$$= \overline{\overline{(A+\overline{B})(AB)}}+\overline{(A+B)(\overline{\overline{A}\overline{B}})}$$

$$= \overline{A\overline{A}B}+\overline{A+B}+\overline{A+\overline{B}} = 1+\overline{A}\overline{B}+\overline{A}B = 1$$

35. **12-13 C3 Boolean Algebra**

$$\overline{A\overline{B}+C}+\overline{\overline{A}+C}+\overline{B\overline{C}} = \overline{A\overline{B}}\,\overline{C}+A\overline{C}+\overline{B}+C \qquad 8$$

$$= (\overline{A}+B)\overline{C}+A\overline{C}+\overline{B}+C = \overline{C}(\overline{A}+B+A)+\overline{B}+C$$

$$= \overline{C}+\overline{B}+C = 1+\overline{B} = 1$$

36. **13-14 C3 Boolean Algebra**

$(AB+\overline{C})(\overline{A}+BC)(A+\overline{B}+C) = (AB\overline{A}+ABBC+\overline{C}\,\overline{A}+\overline{C}BC)(A+\overline{B}+C)$ (1,1,1), (0,0,0)

$= (ABC+\overline{A}\,\overline{C})(A+\overline{B}+C)$

$= (ABCA+ABC\overline{B}+ABCC+\overline{A}\,\overline{C}A+\overline{A}\,\overline{C}\,\overline{B}+\overline{A}\,\overline{C}C)$

$= ABC+\overline{A}\,\overline{B}\,\overline{C}$

$ABC+\overline{A}\,\overline{B}\,\overline{C}=1 \Rightarrow ABC=1 \vee \overline{A}\,\overline{B}\,\overline{C}=1 \Rightarrow (1,1,1)\vee(0,0,0)$

37. **13-14 C3 Boolean Algebra**

$A\overline{B}(AB+\overline{C})+\overline{C}(A+\overline{B}C) = A\overline{B}AB+AB\overline{C}+\overline{C}A+\overline{C}\,\overline{B}C$ $A\overline{C}$

$= AB\overline{C}+A\overline{C} = A\overline{C}(B+1) = A\overline{C}$

38. **14-15 C3 Boolean Algebra**

$(\overline{\overline{A}+B})(B+C)(\overline{A+\overline{C}})(A\overline{B}+BC) = (\overline{\overline{A}\,\overline{B}})(B+C)(\overline{A}\,\overline{\overline{C}})(A\overline{B}+BC)$ 0

$= (A\overline{B}B+A\overline{B}C)(\overline{A}CA\overline{B}+\overline{A}CBC)$

$= A\overline{B}C\overline{A}BC = 0$

39. **14-15 C3 Boolean Algebra**

$\overline{B}(\overline{A}C+B\overline{C})+\overline{C}(AC+\overline{B})+\overline{\overline{B}+C} = (\overline{B}\,\overline{A}C+\overline{B}B\overline{C})+(\overline{C}AC+\overline{C}\,\overline{B})+\overline{\overline{B}}\,\overline{C}$ 5

$= \overline{A}\,\overline{B}C+\overline{B}\,\overline{C}+B\overline{C} = \overline{A}\,\overline{B}C+\overline{C}(B+\overline{B}) = \overline{A}\,\overline{B}C+\overline{C}$

$C=1 \rightarrow \overline{A}\,\overline{B}1+0=\overline{A}\,\overline{B}=1 \rightarrow A=0 \wedge B=0 \rightarrow (0,0,1)$

$C=0 \rightarrow AB0+1=1 \rightarrow A=* \wedge B=* \rightarrow (*,*,0)$ So 5 triples are TRUE.

40. **15-16 C3 Boolean Algebra**

$\overline{A(B+\overline{C})}+\overline{B\overline{C}}+\overline{\overline{A}+B+\overline{C}} = \overline{A}+\overline{B+\overline{C}}+\overline{B}+\overline{\overline{C}}+\overline{\overline{A}}\,\overline{B}\,\overline{\overline{C}}$ $\overline{A}+\overline{B}+C$

$= \overline{A}+\overline{B}\,\overline{\overline{C}}+\overline{B}+C+A\overline{B}C = \overline{A}+\overline{B}C+\overline{B}+C+A\overline{B}C$

$= \overline{A}+\overline{B}(C+1+AC)+C = \overline{A}+\overline{B}+C$

41. **15-16 C3 Boolean Algebra**

$\overline{A\overline{B}}(A+\overline{B}C)+\overline{A}(\overline{B\overline{C}}+A) = (\overline{A}+\overline{B})(A+\overline{B}C)+\overline{A}(\overline{B}+\overline{C}+A)$ 3

$= \overline{A}A+\overline{A}\,\overline{B}C+\overline{B}A+\overline{B}\,\overline{B}C+\overline{A}\,\overline{B}+\overline{A}\,\overline{C}+\overline{A}A = \overline{A}\,\overline{B}C+\overline{B}(A+\overline{A})+\overline{B}C+\overline{A}\,\overline{C}$

$= \overline{B}(\overline{A}C+1+C)+\overline{A}\,\overline{C} = \overline{B}+\overline{A}\,\overline{C}$

So $B=1 \wedge \overline{A}\,\overline{C}=0$. Thus 3 triples make it FALSE: (1,1,1), (1,1,0), (0,1,1).